Hongpei Li

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Education

Northwestern University 2026 - 2031 (Expected)
Ph.D. in Industrial Engineering and Management Sciences (IEMS) Evanston, IL, USA
Advisor: Prof. Zikai Xiong.

Shanghai University of Finance and Economics 2021 - 2025
Bachelor of Engineering, Pilot Class of Interdisciplinary Sciences Shanghai, China

Research Interests

Large-Scale Optimization on AI-Era Computing Hardware; Intersection of AI and Optimization

Publications

Large-Scale Optimization

- A Restarted Primal-Dual Hybrid Conjugate Gradient Method for Large-Scale Quadratic Programming
  Y. Huang, W. Zhang, **H. Li**, D. Ge, H. Liu, and Y. Ye (2025). **INFORMS Journal on Computing**. [Paper] [Code]
- PDHCG-II: An Enhanced Version of PDHCG for Large-Scale Convex QP
  H. Li, Y. Huang, H. Liu, D. Ge, and Y. Ye (2026). *arXiv preprint.* [Paper] [Code]
- A Scalable First-Order Method for Large-Scale Competitive Market Equilibrium Computation
  H. Liu, Y. Huang, **H. Li**, D. Ge, and Y. Ye (2025). *arXiv preprint.* [Paper] [Code]
- D-PDLP: Scaling PDLP to Distributed Multi-GPU Systems
  H. Li, Y. Huang, H. Liu, D. Ge, and Y. Ye (2026). *arXiv preprint.* [Paper] [Code]
- A Curvature-Aware Rank-Adaptive Distributed Augmented-Lagrangian Solver for Large-Scale SDPs
  H. Li, H. Liu, D. Ge, and Y. Ye (2026). *arXiv preprint.* [Paper] [Code]
- GPU-Accelerated Conic Quadratic Programming with Local Linear Convergence under Strict Complementarity
  H. Li, Y. Huang, H. Liu, D. Ge, and Y. Ye (2026). *arXiv preprint.* [Paper] [Code]

AI and Optimization

- A Learning and Searching Method for Integrated Process Planning and Scheduling
  H. Li, H. Zhang, Z. He, J. Hu, and H. Yuan (2026). **Production and Operations Management**. [Paper] [Code]
- BenLOC: A Benchmark for Learning to Configure MIP Optimizers
  H. Li, Z. He, Y. Wang, S. Pu, Q. Deng, W. Tu, and D. Ge (2025). *arXiv preprint.* [Paper] [Code]
- FMIP: Joint Continuous-Integer Flow for Mixed-Integer Linear Programming
  H. Li, H. Yuan, H. Zhang, J. Lin, D. Ge, M. Wang, and Y. Ye (2026). **ICLR 2026**. [Paper] [Code]

Machine Learning Systems

- OptPipe: Memory- and Scheduling-Optimized Pipeline Parallelism for LLM Training
  H. Li, H. Zhang, H. Liu, D. Ge, and Y. Ye (2025). *arXiv preprint.* [Paper]
- OSDN: Improving Delta Rule with Provable Online Preconditioning in Linear Attention
  C. Zhou, **H. Li**, Y. Liu, J. Lin, D. Ge, and Y. Ye (2026). *arXiv preprint.* [Paper]

Software

- **PDHCG**: GPU-accelerated primal-dual hybrid conjugate gradient solvers for quadratic and conic quadratic programming.
- **CARDAL**: A multi-GPU, rank-adaptive augmented Lagrangian solver for large-scale semidefinite programming.
- **D-PDLP**: A distributed primal-dual first-order solver for large-scale linear programming across multiple GPUs.

Work Experience

QingTian Lab, Cardinal Operations Feb. 2025 - Jun. 2026
LLM & OPT Researcher
Developed GPU-accelerated optimization solvers for COPT and conducted research on large language model inference.

Teaching Experience

Shanghai University of Finance and Economics

- Advanced Operations Research (TA) Spring 2024
- Object-Oriented Analysis & Design (TA) Fall 2023
- Programming Peer Tutor Fall 2022

Honors & Awards

- Honor Bachelor Degree, 2025. **(Top 5% in Major)**
- Honor of Excellence Graduation Thesis, 2025. **(Top 3% in Major)**
- Shanghai Municipal Bronze Award in the Internet+ University Student Innovation and Entrepreneurship Competition. **(0.13%)**